

ORIGINAL RESEARCH

A novel approach to partial coverage in wireless sensor networks via the roman dominating set

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Abstract

One major challenge in deploying wireless sensor networks (WSN) in real-world applications is minimising the energy consumption by the sensors while maintaining the coverage of the monitored field. However, many applications do not need full coverage of the monitored area all the time, which can help us reduce the network's energy consumption. One approach to exploit this property is to set up a sleep/wake-up schedule for each node such that no redundant nodes are active in an area of coverage simultaneously. This will allow the existence of monitoring holes in a controlled manner, which the authors call *partial coverage*. In this study, the partial coverage condition is imposed by constructing a Roman Dominating Set of awake nodes in the network. Roman domination is a method for colouring a graph's vertices with three labels (0, 1, 2), such that all the vertices with label 0 have an adjacent vertex with label 2, while the sum of the labels of nodes is minimised. Based on this formulation, a simple greedy algorithm is proposed to construct such a structure supported by three theorems. Furthermore, the performance of the authors' proposal in different scenarios is evaluated.

1 | INTRODUCTION

A wireless sensor network (WSN) is an ad hoc network consisting of sensors where each sensor has the capability of sensing or processing data and communicating it to other nodes. The main function of WSN is to gather data from the surrounding area and communicating the data to a gateway for further processing. WSNs have a wide range of applications, such as forestry fire monitoring [1–4], environmental monitoring [5, 6], monitoring water quality [7], among many others [8–12]. However, since usually the nodes are equipped with cheap non-replaceable batteries, minimising energy consumption while having an acceptable field coverage is the main challenge in WSNs [13–19].

1.1 | Problem statement and our contributions

The concept of coverage mentioned above refers to the extent of monitoring the area of interest, which depends on the

placement of nodes and their sensors range. Different applications require different levels of coverage. There are four main types of covering an area explored in the literature as follows:

1. *Complete coverage*: In some applications, the entire area needs to be monitored all the time, meaning that at least one node should be active within every region of coverage [20].
2. *Border coverage*: Sometimes, it is only needed to put sensors along the border of an area to monitor crossing the border. Therefore, observing the inner area is not required [21].
3. *Point of Interest (POI) coverage*: Some applications only require keeping track of some specific points of interest [22].
4. *Partial coverage*: Many applications do not require the entire area to be covered all the time [13–19].

In this study, we consider the case of partial coverage, when monitoring the entire area is not necessary all the time

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but may be required at some moments. For example, in the forest fire detection application, the whole region may need to be covered in hot seasons, while it may not be necessary in cold seasons [19]. Therefore, more sensors can be set to the sleep mode to minimise energy consumption and set back to the active mode whenever necessary. Usually, the sensors in the network are deployed in such a way that multiple nodes cover each region. Thus, one way to improve survivability is to set only one node in each area awake while other nodes covering the same region are put to sleep. Nodes in the sleeping state stop radio transmission and sensing, saving a significant part of their energy. Moreover, in the case of partial coverage, one solution is to let a region uncovered, as the complete coverage of the area may not be needed as mentioned before.

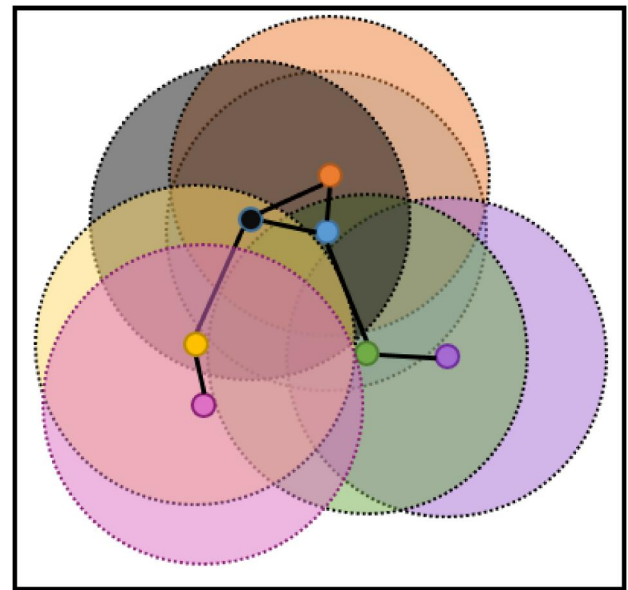
The idea of setting the nodes with correlation into sleep mode has been implemented in several WSN partial coverage solutions [13–19]. However, the placement of uncovered regions within the entire area has not gained much attention. Although the percentage of coverage is the most common measure for evaluating partial coverage methods, the distribution of coverage holes within the area is also important, as a large uncovered area is not acceptable. In contrast to previous studies, our method ensures that the uncovered regions are sparse and are surrounded with covered regions. We use the Roman dominating set as the core idea of our method, as in this method, nodes are set to sleep only if at least one neighbouring node is awake and is covering their region, or the surrounding areas are all covered. The Roman dominating set ensures that the uncovered regions are distributed all over the coverage area and no big region is left uncovered. Whenever complete coverage is needed, one sensor in the holes can be set back to the active mode.

Based on these concepts, we formulate the partial coverage problem via a graph-theoretic approach as follows. A WSN is modelled with a graph $G(V, E)$, where V is the set of all nodes. The edge (u, v) exists if the two sensors are within the same sensing range. We assume the binary sensing model for the sensors. In the binary sensing model, a node can sense the events within a certain range around itself with the probability of one and cannot sense any events outside of its sensing range. Another sensing model for nodes in WSN is a probabilistic sensing model, where the probability of a node sensing an event decreases with respect to the distance between the node and the event [23]. We aim to select a number of nodes to be awake and set the rest of the nodes to sleep in a way to satisfy the following condition. A node may be in the sleep mode if another node within the same sensing range is awake or at least one awake node is available at every adjacent sensing range. In other words, all the areas within a region should be covered except those whose neighbouring regions are covered.

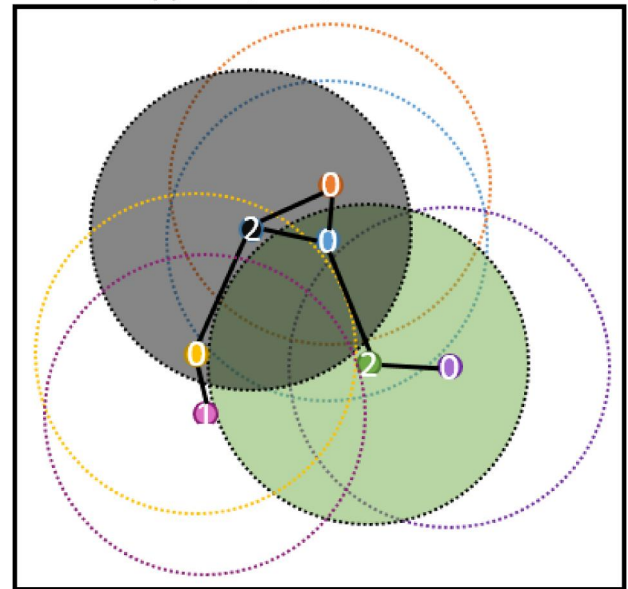
To model such conditions, we use the notion of the Roman dominating function. The Roman dominating function partitions the set of the graph nodes into three subsets as $V = \{V_0 \cup V_1 \cup V_2\}$, such that all the vertices in V_0 should be adjacent to at least one vertex from the subset V_2 . The

vertices from the subsets V_0 and V_1 are set to sleep, and the vertices from the subset V_2 are set to be awake. The vertices from the subset V_1 are generally set to sleep unless we need a higher percentage of coverage of the area.

As demonstrated in the example of Figure 1a, there is an edge between the nodes, which are within each other's sensing range. Labels assigned to the nodes of this network by a possible Roman dominating function are shown in the Figure 1b. Each node marked with the number 0 belongs to the subset V_0 , nodes marked with the number 2 belong to V_2 , and the node labelled 1 belongs to V_1 . As shown, each node in V_0 has a neighbour in V_2 . We will experimentally show that the nodes in V_2 cover a larger area than another subset of the same number of randomly selected nodes.



(a) Sensing coverage of all the nodes



(b) Sensing coverage of the nodes with label 2

FIGURE 1 A wireless sensor networks and its related graph

In summary, the contributions of this study are as follows:

- We consider a partial coverage situation, where it is possible to have single holes in the coverage. A hole in the coverage is a single area not covered, where all its adjacent areas are covered. We propose to model and formulate such a scenario by constructing a Roman dominating set.
- We propose a greedy algorithm for obtaining the minimum Roman dominating set with low computational complexity. Our greedy algorithm is built upon important observations made rigorous via three structural theorems. These theorems state necessary conditions for constructing minimum Roman dominating sets. The first theorem states that a node with the label of 1 does not have more than one neighbour with the same label in a minimum Roman dominating function. According to the first and second theorems, a node with the label of 1 does not have a neighbour labelled 2 and does not have more than one neighbour with the label of 1. The third theorem states that no 0-labelled vertex can have more than two neighbours with the label of 1.
- Via extensive simulations, we evaluate our algorithm via different scenarios, which show the benefits of our proposal.

1.2 | Related work

Several different solutions have been proposed to address the problem of energy constraint in wireless sensor networks based on various methods regarding the energy management of nodes. One common approach is to implement a sleep/wake-up schedule to maintain the energy of sleeping nodes. If the entire area is to be covered, the scheduling problem can be represented as a *minimum dominating set problem* [24, 25].

In another case, if monitoring the entire area is not necessary, we can relax the coverage constraint and let an area uncovered if all its surrounding areas are covered. We represent this problem as a minimum Roman dominating set.

Guha and Khuller [26] used the dominating set algorithm for controlling the sleep/wake-up schedule of sensors for the first time. They introduced approximation algorithms with the complexity of $O(\log n)$, where n is the number of nodes in the graph.

Next, Islam et al. [27] introduced the concept of the *domatic partition problem* in unit disk graphs, which is producing the maximum number of disjoint dominating sets. They proposed a distributed algorithm to construct connected dominating sets without the direct use of a maximal independent set (MIS). In Ref. [28], the connectivity of a connected dominating set (CDS) is modified using a non-trivial potential function. The proposed algorithm is based on a decomposition of a 2-connected graph into 3-connected components, which is commonly known as Tutte's decomposition. In Ref. [24], the sensors are assumed to have different initial energy states. A number of local search algorithms for finding the domatic partitions are proposed in Ref. [24].

The minimum Roman dominating set problem is motivated initially by the defence strategy used by the Roman Empire.

Stewart [29] introduced the notion of Roman dominating function for graphs in the article: 'Defend the Roman empire!'. In Ref. [29], it is shown that the problem of finding a minimum Roman dominating set is NP-hard. Most of the studies on this problem and its different versions have been focussed on finding tighter bounds for the associated domination numbers (see, e.g. Burger et al. [30] and Chambers et al. [31]). Meanwhile, some sub-optimal algorithms exist for several unique graphs [32]. In Ref. [33], a linear-time algorithm is presented for the Roman domination problem in the unit disc graphs, which is widely popular in modelling homogeneous WSNs. In Ref. [34], a generalisation of the Roman domination problem, called connected strong k -Roman dominating set, is proposed, and an approximation algorithm is presented for the unit ball graphs. Unit ball graphs are used to model three-dimensional wireless sensor networks. Generally speaking, unlike the classic case of dominating sets, algorithms and applications regarding Roman dominating sets have not attracted much attention among researchers.

Finally, the results of the proposed method are compared to two other graph theoretical WSN partial coverage methods. The CCDS method [35] is a partial coverage algorithm designed to prolong the network lifetime. The first phase of this method is constructing a MIS. A MIS is also a dominating set, and in order for it to be connected, the algorithm adds connecting nodes to the active nodes set, obtaining the final connected dominating set. The SCCS method is also a CDS-based method, which chooses the nodes with the least spatial correlation with other nodes. To this end, the algorithm chooses one node in each clique in the complement graph of the sensor network.

1.3 | Paper structure

The study is organised as follows. In Section 2, we overview the notations and state the problem being addressed in this study. In Section 3, we state and prove three structural theorems regarding the minimum Roman dominating function, which are critical in our proposed algorithm design. In Section 4, we first review a simple greedy algorithm, and then a novel improved alternative is proposed. In Section 5, the results of the two algorithms are compared with a random method and two other graph theoretical WSN partial coverage methods ([35, 36]) via extensive simulations. Finally, Section 6 concludes the study.

2 | NOTATIONS AND THE PROBLEM STATEMENT

This section first reviews some definitions and notations used in the study and then states the main problem being addressed.

2.1 | Definitions and notations

For a vertex $v \in V$ in the graph $G(V, E)$, the set of all adjacent nodes to v is called the open neighbourhood of that

vertex and is denoted by $N(v)$. The closed neighbourhood of v is also defined as: $N[v] \triangleq N(v) + \{v\}$. The size of the set $N(v)$ is called the degree of v and is denoted by $d(v)$. We may also define a function named $d^2 : V \rightarrow \mathbb{N} + \{0\}$, which maps each vertex to sum of its neighbouring vertexes' degree, such as $d^2(u) = \sum_{v \in N(u)} d(v)$.

Neighbouring nodes of a vertex v , which have not been labelled, form a set called unlabelled neighbours of v and are denoted by $NU(v)$, where the size of this set is called the unlabelled degree of v and is shown as $du(v)$.

For every two nodes $u, v \in V$, $d(u, v)$ denotes the number of edges in the shortest path connecting them and is called the distance between u and v . A 2-packing (P) of a graph is a subset of the graph's nodes such that for every $u, v \in P$, $d(u, v) \geq 3$.

2.2 | Problem statement

The wireless sensor network is modelled as an undirected graph, $G(V, E)$, where V is the set of nodes in the network. Also, $(v_1, v_2) \in E$ denotes that the nodes v_1, v_2 are within each other's sensing range. Here, we aim to find a subset of nodes covering the largest area, compared to other subsets with the same size. Our goal is to set a sleep/wake-up schedule for sensor nodes so that each node is at most within the distance of two hops from an active node, such that the number of active nodes is minimised.

Roman domination is a function on graph $G(V, E)$ defined as: $f : V \rightarrow \{0, 1, 2\}$, satisfying the condition that every vertex u with $f(u) = 0$ should have at least one neighbour v such that $f(v) = 2$ [29]. This function is called an RD function. The weight of a RD function is defined as $w_G(f) = \sum_{v \in V} f(v)$.

For a graph G , we denote γ_R^G as the minimum weight of a Roman domination function. Therefore, we can write

$$\gamma_R^G = \min\{w_G(f) \mid f \text{ is an RD-function on } G\} \quad (1)$$

We call the corresponding RD function the minimum RD function and denote it by MRD function. We aim at building an algorithm that finds an RD function in a way that $w_G(f)$ is as close as possible to γ_R^G .

In order to do this, in the next section, we state three features of the MRD function in the form of three theorems with their proofs. In Section 4, we construct an algorithm to find an RD function based on features stated in these theorems. The simple greedy algorithm is modified such that no vertex with the label 1 is adjacent to another vertex with the label of 1 or 2. Furthermore, no vertex with the label 0 is adjacent to more than one vertex with the label 1. The results of our proposed algorithm are compared to the results of the simple greedy algorithm in Section 5.

3 | FEATURES OF AN MRD FUNCTION

As stated before, an RD function is a function $f : V \rightarrow \{0, 1, 2\}$, such that every vertex u , for which $f(u) = 0$, is adjacent to at least one vertex v such that $f(v) = 2$. We call the set of all possible RD functions, RDF. An MRD function is an RD function that satisfies the following condition:

$$\text{MRD-function} \in \text{RDF} \leftrightarrow \forall f \in \text{RDF}, \quad w_G(f) < w_G(\text{MRD-function}) \quad (2)$$

We also define the sets V_0, V_1, V_2 as follows

$$V_i = \{v \in V(G) \mid \text{MRD}(v) = i\} \quad (3)$$

for $i = 0, 1, 2$. Now, we are ready to present the following theorems for every MRD function.

Theorem 1 Every vertex $v \in V_1$ has at most one neighbour from the set V_1 .

Proof. Suppose that there is $v_1, v_2, v_3 \in V_1$, such as

$$(v_1, v_2), (v_1, v_3) \in E(G) \quad (4)$$

Then, there is a function $g(\cdot) \in \text{RDF}$, such as:

$$\forall v \in V(G), \quad v \neq v_1, v_2, v_3 \rightarrow g(v) = \text{MRD}(v), \quad g(v_1) = 2, g(v_2) = 0, g(v_3) = 0 \quad (5)$$

Then we will have

$$\begin{aligned} w_G(f) &= \sum_{v \in V} f(v) \\ &= \sum_{\substack{v \in V \\ v \neq v_1, v_2, v_3}} f(v) + f(v_1) + f(v_2) + f(v_3) \\ &= \sum_{\substack{v \in V \\ v \neq v_1, v_2, v_3}} f(v) + 2 + 0 + 0 \\ &= \sum_{\substack{v \in V \\ v \neq v_1, v_2, v_3}} f(v) + 2 \end{aligned} \quad (6)$$

whereas,

$$\begin{aligned} \gamma_R^G &= \sum_{v \in V} \text{MRD-function}(v) \\ &= \sum_{\substack{v \in V \\ v \neq v_1, v_2, v_3}} \text{MRD}(v) + \text{MRD}(v_1) \\ &\quad + \text{MRD}(v_2) + \text{MRD}(v_3) \\ &= \sum_{\substack{v \in V \\ v \neq v_1, v_2, v_3}} \text{MRD}(v) + 1 + 1 + 1 \\ &= \sum_{\substack{v \in V \\ v \neq v_1, v_2, v_3}} \text{MRD}(v) + 3 \end{aligned} \quad (7)$$

and as stated before,

$$\sum_{\substack{v \in V \\ v \neq v_1, v_2, v_3}} f(v) = \sum_{\substack{v \in V \\ v \neq v_1, v_2, v_3}} MRD(v) \quad (8)$$

so

$$w_G(f) < \gamma_R^G \quad (9)$$

and this result contradicts the definition of γ_R^G .

Theorem 2 No vertex from the set V_1 is adjacent to a vertex from the set V_2 .

Proof. Suppose that there is $v_1, v_2 \in V_1$, such that

$$(v_1, v_2) \in E(G) \quad (10)$$

and suppose that $MRD(v_1) = 1$, $MRD(v_2) = 2$. Define a function $f \in \text{RDF}$, such as $\forall v \in V(G)$, $v \neq v_1 \rightarrow f(v) = MRD(v)$, $f(v_1) = 0$ then,

$$\begin{aligned} w_G(f) &= \sum_{v \in V} f(v) = \sum_{\substack{v \in V \\ v \neq v_1}} f(v) + f(v_1) \\ &= \sum_{\substack{v \in V \\ v \neq v_1}} f(v) + 0 = \sum_{\substack{v \in V \\ v \neq v_1}} f(v) \end{aligned} \quad (11)$$

whereas,

$$\begin{aligned} \gamma_R^G &= \sum_{v \in V} MRD(v) = \sum_{\substack{v \in V \\ v \neq v_1}} MRD(v) + MRD(v_1) \\ &= \sum_{\substack{v \in V \\ v \neq v_1, v_2, v_3}} MRD(v) + 1 \end{aligned} \quad (12)$$

and as stated before,

$$\sum_{\substack{v \in V \\ v \neq v_1}} f(v) = \sum_{\substack{v \in V \\ v \neq v_1}} MRD(v) \quad (13)$$

so

$$w_G(f) < \gamma_R^G \quad (14)$$

and this result contradicts the definition of γ_R^G .

Theorem 3 Every vertex from the set V_0 has at most two neighbours from the set V_1 .

Proof. Suppose that there are $v_1, v_2, v_3, v_4 \in V_1$, such that

$$(v_1, v_2), (v_1, v_3), (v_1, v_4) \in E(G) \quad (15)$$

where

$$MRD(v_1) = 0, MRD(v_2) = MRD(v_3) = MRD(v_4) = 1 \quad (16)$$

Then, define a function $f \in \text{RDF}$, such as,

$$\begin{aligned} \forall v \in V(G), v \neq v_1, v_2, v_3 \rightarrow \\ f(v) = MRD(v), f(v_1) = 2, f(v_2) = f(v_3) = f(v_4) = 0 \end{aligned} \quad (17)$$

Then, we can proceed as follows:

$$\begin{aligned} w_G(f) &= \sum_{v \in V} f(v) \\ &= \sum_{\substack{v \in V \\ v \neq v_1, v_2, v_3}} f(v) + \sum_{v \in \{v_1, \dots, v_4\}} f(v) \\ &= \sum_{\substack{v \in V \\ v \neq v_1, v_2, v_3}} f(v) + 2 + 0 + 0 + 0 \\ &= \sum_{\substack{v \in V \\ v \neq v_1, v_2, v_3}} f(v) + 2 \end{aligned} \quad (18)$$

and as stated before,

$$\sum_{\substack{v \in V \\ v \neq v_1, v_2, v_3}} f(v) = \sum_{\substack{v \in V \\ v \neq v_1, v_2, v_3}} MRD(v) \quad (19)$$

so

$$w_G(f) < \gamma_R^G \quad (20)$$

and this result contradicts the definition of γ_R^G .

Based on the features provided above in the form of three theorems, we describe our proposed algorithm in the next section.

4 | THE PROPOSED ALGORITHM

In this section, our primary goal is to develop an efficient algorithm that labels the vertices of a given graph $G(V, E)$, with labels from the set $L = \{0, 1, 2\}$. Such that every vertex with label 0 has at least one neighbour with label 2, in a way that the sum of all vertices' labels is minimised.

We first introduce a simple greedy algorithm for computing the Roman dominating number. Then, we use the above-stated theorems to modify this algorithm which improves the performance. This algorithm is based on the greedy algorithm for the minimum dominating set problem [37–39].

Algorithm 1 The Simple Greedy algorithm

```

1: procedure GREEDY ALGORITHM( $G$ )
2:   while  $V(G)$  not empty do
3:     for all  $v \in V(G)$  do
4:       Calculate  $du(v)$ 
5:       if  $du(v) = 0$  then
6:          $f(v) = 1$ 
7:         remove  $v$  from  $V(G)$ 
8:       end if
9:     end for
10:     $u = \arg\text{Max}\{du(u)\}$ 
11:     $f(u) = 2$ 
12:    remove  $v$  from  $V$ 
13:    for all  $j \in N_V[u]$  do:
14:       $f(j) = 0$ 
15:      remove  $j$  from  $V(G)$ 
16:    end for
17:  end while
18: end procedure

```

In the following, we explain the main intuition behind the simple greedy algorithm, demonstrated in Algorithm 1.

In order to calculate the minimum Roman dominating number of a graph, one needs to construct a Roman domination function, $f(u) = i \in \{0, 1, 2\}$, such that $\sum_u f(u)$ is minimised. Intuitively speaking, in order to minimise the Roman dominating number, we always prefer to label a vertex as 0, rather than 1 or 2, as long as a neighbouring vertex with label 2 dominates it. If a vertex is not dominated and is not able to dominate any other vertices, it cannot be assigned the label 0, and it should not be assigned the label 2. Therefore, it is labelled as 1. We prefer to assign the label 2 to vertices which can dominate the largest number of other vertices. In other words, vertices with the largest number of neighbouring unlabelled vertices are in priority to be labelled with 2.

Therefore, in every loop of the algorithm, the vertex with the largest $du(u)$ is labelled 2, and all the unlabelled neighbouring vertices, which are now dominated, are labelled with the label 0. Eventually, vertices which all of their neighbours are labelled as 0 are assigned the label 1, since they cannot be dominated or dominate any other vertices.

The first algorithm is constructed as follows: in every loop, $du(v)$ is calculated for each vertex. Vertices with $du(v) = 0$ are not dominated and thus, we set $f(v) = 1$. Afterwards, all the non-labelled vertices are arranged in order of the unlabelled degree. The vertex with the largest unlabelled degree is labelled as 2, and all its neighbouring vertices are labelled 0. The labelling function constructed within this algorithm is a Roman dominating function because assigning the label 0 to any vertex is subjected to assigning the label 2 to one of its neighbours beforehand. Hence, in the end, all the vertices whose label is 0 are dominated, satisfying the necessary and sufficient condition for a Roman dominating function.

At the beginning of each loop, those vertices not having any adjacent unlabelled vertices are necessarily surrounded by

0-labelled vertices. This statement can be easily proved by contradiction. Assuming that such a vertex has a neighbouring 2-labelled vertex, this vertex should be labelled in a previous algorithm loop. Thus, in the next step, all its neighbours are labelled 0, and none of its neighbours are left unlabelled for the following loops. Therefore, one can make sure that a vertex with no unlabelled neighbours at the beginning of the algorithm should be labelled 1.

Algorithm 1 explained above was the most straightforward greedy algorithm for finding the Roman dominating number for any given graph. In what follows, we employ previously stated theorems regarding the features of a minimum Roman dominating function, to improve the performance of the above algorithm. Moreover, due to practical reasons, we aim to decrease the number of vertices labelled as 1, while the total Roman dominating number is the same. For example, we prefer to have a vertex labelled 2 with a neighbour labelling 0, rather than two adjacent vertices with label 1, provided that this change will not affect the label of the remaining nodes. Based on this attitude and the previously stated theorems, the following modifications can be applied to the simple greedy algorithm.

Algorithm 2 The Improved Greedy Algorithm

```

1: procedure IMPROVED ALGORITHM( $G, u, l, p$ )
2:   while  $V(G)$  not empty do
3:     for all  $v \in V(G)$  do
4:       Calculate  $du(v)$ 
5:       if  $du(v) = 0$  then
6:         if  $\{V_1, v\}$  is 2-packing then
7:            $f(v) = 1$ 
8:           add  $v$  to  $V_1$ 
9:           remove  $v$  from  $V(G)$ 
10:        else
11:          for all  $u \in V_1$  do
12:            if  $(v, u) \in E$  then
13:               $f(u) = 2$ 
14:               $f(v) = 0$ 
15:              remove  $u, v$  from  $V(G)$ 
16:              for all  $j \in N[u]$  do
17:                 $f(j) = 0$ 
18:                remove  $j$  from  $V(G)$ 
19:              break
20:            end for
21:          else if  $(u, w), (v, w) \in E$  then
22:             $f(w) = 2, f(v) = 0, f(u) = 0$ 
23:          end if
24:        end for
25:      end if
26:    end if
27:  end for
28:   $\{u\} = \arg\text{Max}\{du(u)\}$ 
29:   $u = \arg\text{Max}\{\text{sum}(du(V, u))\}$ 
30:   $f(u) = 2$ 
31:  remove  $v$  from  $V$ 
32:  for all  $j \in N_V[u]$  do:
33:     $f(j) = 0$ 

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34:     remove  $j$  from  $V(G)$ 
35:   end for
36: end while
37: end procedure

```

In the modified algorithm, same as the first one, we begin with labelling the vertices with no unlabelled neighbours left and label them as 1. According to Theorem 1, in a minimum Roman dominating function, no vertex with label 1 is adjacent to more than one vertex with label 1. As we prefer to decrease the number of vertices labelled as 1, we do not let two 1-labelled vertices adjacent to each other. Instead, we label one of them as 2 and the other one as 0. The vertex labelled 0 is dominated because of the adjacent vertex with label 2, so the dominating function is still valid, and the domination number has not been changed. Therefore, if a vertex with no unlabelled neighbouring vertex has an adjacent vertex with label 1, it is labelled as 2, and the label of that neighbouring vertex is changed to 0; otherwise, it is labelled as 1.

Moreover, according to Theorem 3, we have to check that no vertex with label 0 is adjacent to more than two vertices with label 1. As we aim to reduce the number of 1-labelled vertices, we will not let a vertex with label 0 have more than one 1-labelled neighbour. In other words, the subset of 1-labelled vertices should form a 2-packing in the graph.

Therefore, our proposed algorithm is constructed as follows. At the beginning of each loop, vertices with no unlabelled neighbours are identified. They will then be checked to see if they form a 2-packing with the subset of other vertices with label 1 or not; if so, it will be assigned the label 1 and added to the subset of 1-labelled vertices. If the new vertex is adjacent to another 1-labelled vertex, it will be labelled as 2, and its neighbours will be assigned the label 0. If there is another 1-labelled node within the distance 2 of this node, it means that a 0-labelled vertex is adjacent to both of them (as we have already proved that no vertex with label 2 is adjacent to a 1-labelled vertex). In this case, the middle 0-labelled vertex will be relabelled as 2, and the other two 1-labelled vertices are assigned the label 0.

Afterwards, all the unlabelled nodes are arranged in order of their unlabelled degree. If there is only one vertex with the maximum unlabelled degree, it will be labelled as 2, and all its unlabelled neighbours will be assigned as 0. If there are more than one vertices with the maximum degree, the one with the least sum of the neighbour degrees will be chosen for assigning the label 2. Unlike the first algorithm, where if more than one vertex had the maximum unlabelled degree, one would be selected randomly.

5 | EXPERIMENTAL RESULTS

In order to evaluate and compare the two previously stated algorithms, we construct two testing scenarios. In the first scenario, we compare both algorithms' results with the optimal Roman dominating number for small graphs. In the

second scenario, we compare the results of the algorithms for Unit Disc graphs to see the improvement in the second algorithm's performance and compare the coverage of the algorithms with the case when the active nodes are selected randomly.

For the first scenario, we have constructed a dataset of all graphs with five to eight vertices, 20,000 graphs with nine vertices, and 20,000 graphs with 10 vertices. Then, the minimum Roman dominating number and its corresponding Roman dominating function are saved for each graph. To build this dataset, all possible Roman dominating functions were calculated for each graph, and the one with the minimum Roman dominating number was considered.

To examine the algorithms with this dataset, we have tested each algorithm for each of the dataset graphs. We have computed the error percentage of each algorithm for graphs with different sizes. The error percentage shows that for what percent of test graphs, the Roman dominating number equals the minimum Roman dominating number of that graph. The results are shown in Table 1.

As shown in Table 1, the error increases with the increase of the graph size. The error percentage of Algorithm 2 is slightly bigger than the error percentage of Algorithm 1 for graphs with sizes 9 and 10.

In the second experiment, the proposed algorithms are tested on Unit Disc graphs, broadly accepted models for wireless sensor networks. A Unit Disc graph is a graph whose vertices represent points in the Euclidean plane, each with a circle of coverage around them. There is an edge between two nodes only if one is within the other one's circle. Unit Disc graphs are commonly used for modelling wireless sensor networks, as the nodes in WSN are also implemented on an area and can communicate if they are within a fixed range of each other. Hence, the graph vertices represent wireless sensor nodes, and an edge between two vertices means that the two sensor nodes are within each other's sensing range, therefore have a considerable spatial correlation [33].

In this experiment, we assume that there are two types of sensor nodes with different coverage ranges and, therefore, different radius of coverage circles [40–42]. One type with a smaller coverage area represents cheaper sensors with less sensing/communication capabilities. In comparison, the type with a larger coverage area represents more powerful nodes with more expensive hardware. This heterogeneous structure covers a wide range of sensor network applications. In order to generate this model, random points were generated with uniform distribution in a 20×20 square area, 10% of which had a greater coverage radius (i.e. type 2 nodes). Using this method, a number of pseudo-random

TABLE 1 Errors of Algorithms 1 and 2 for graphs of specific sizes

Alg/Error	5v	6v	7v	8v	9v	10v
Alg_1 (%)	0.00	0.00	0.35	1.18	3.07	4.95
Alg_2 (%)	0.00	0.00	0.35	1.18	3.18	5.10

unit disc graphs with 100–1500 vertices were generated. The minimum Roman dominating set for each graph is calculated using both algorithms. The area covered by the active nodes was compared in both cases when active nodes are chosen using the algorithms and chosen randomly. The results of this experiment is shown in Figures 2 and 3.

Figure 2 shows the percentage of coverage when only nodes with the label 2 are active, with respect to the area covered when all the nodes are active. However, Figure 3 shows the percentage of coverage when the nodes with the label 2 and 1 are active, with respect to the area covered when all the nodes are active.

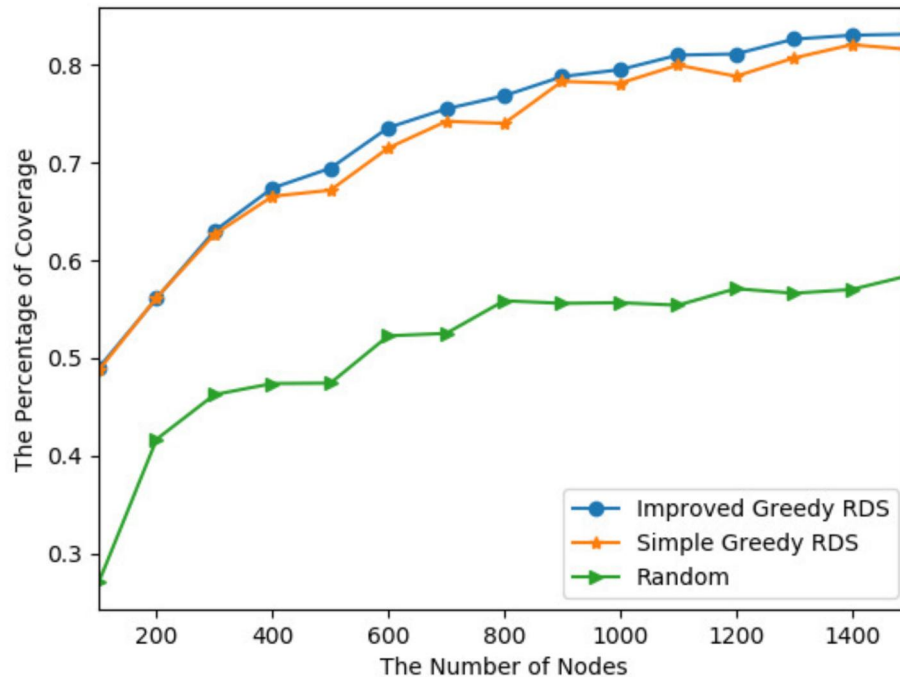


FIGURE 2 The area coverage percentage for different algorithms (only 2-labelled nodes are active)

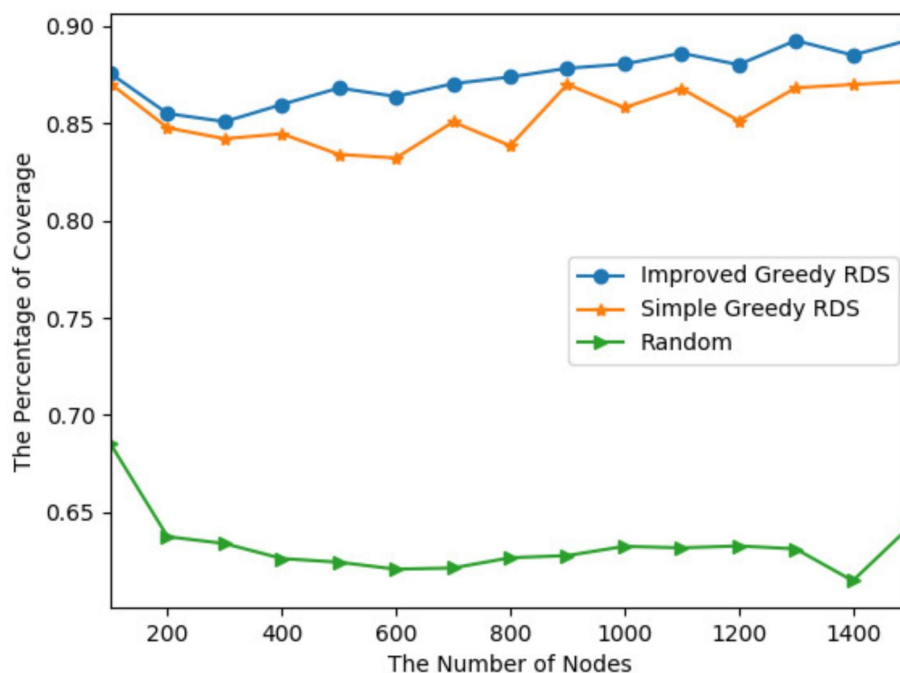


FIGURE 3 The area coverage percentage for different algorithms (both 2-labelled and 1-labelled nodes are active)

Figure 4 shows the coverage of awake nodes with different percentages of nodes with a larger range. When the value of the x -axis is 0, all the nodes have a coverage radius of 1. When the value of the x -axis is 1, all the nodes have a coverage radius of 2.

Figure 5 shows the percentage of coverage with respect to the total coverage with different ranges for the node with the larger coverage range.

We have also compared our proposed method to two other state-of-the-art partial coverage methods ([35, 36]). Both methods find a connected dominating set for the wireless sensor network. The first method first finds a maximum independent set (MIS) within the network and constructs a connected dominating set based on the respective MIS [35]. We refer to this method as the CCDS method. The second

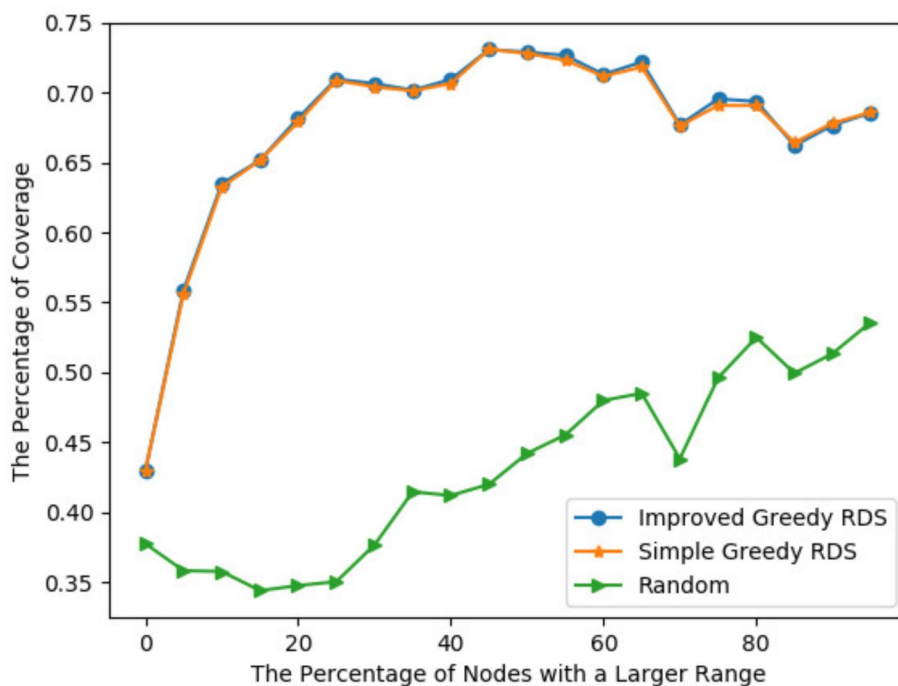


FIGURE 4 The area coverage percentage for different algorithms with respect to the percentage of nodes with the larger range

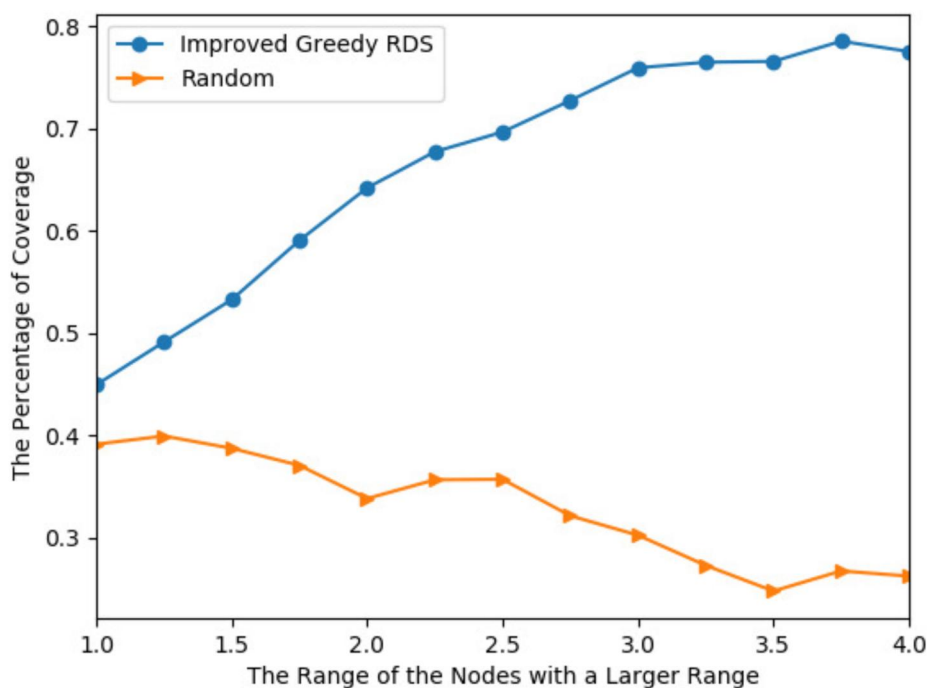


FIGURE 5 The area coverage percentage for different algorithms with different ranges for the nodes with the larger range

algorithm uses a clique construction method to minimise the spatial correlation between the nodes in CDS [36]. This method is referred to as the SCCS method.

Figure 6 shows the percentage of coverage normalised by active nodes for our proposed method versus CCDS and SCCS methods for networks with different sizes, which shows the benefit of our proposed method over them.

In another experiment, we have implemented each method 50 times for a random unit disc graph of size 1200 and generated a scatter plot, where the x -axis is the number of active nodes and the y -axis is the percentage of covered area by the active nodes in each method. The results of this experiment is shown in Figure 7. The scatter plot in Figure 7 shows that the SCCS method generally has a lower performance

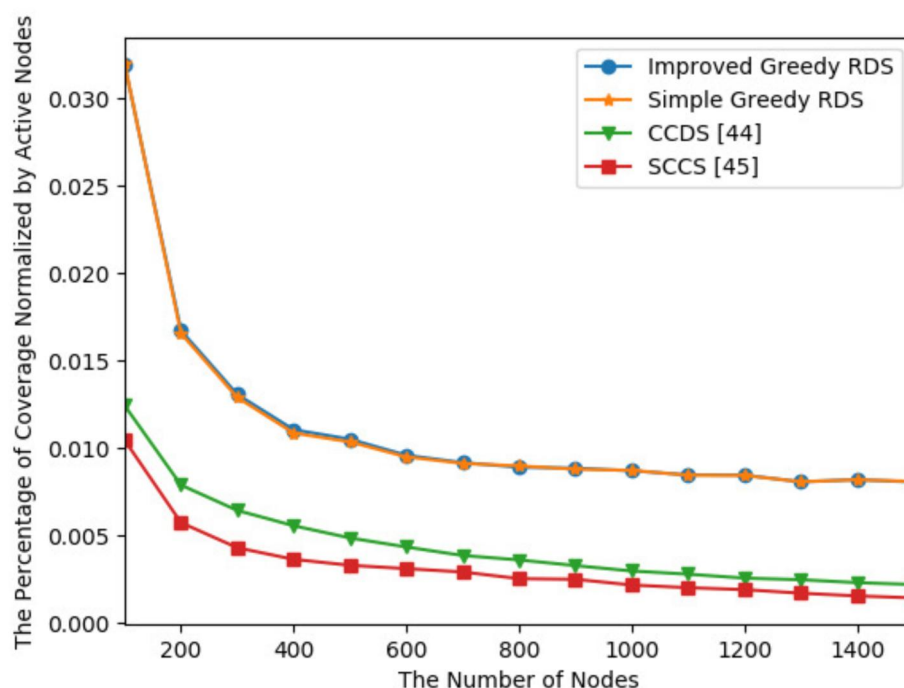


FIGURE 6 The area coverage percentage normalised by the number of active nodes for different algorithms

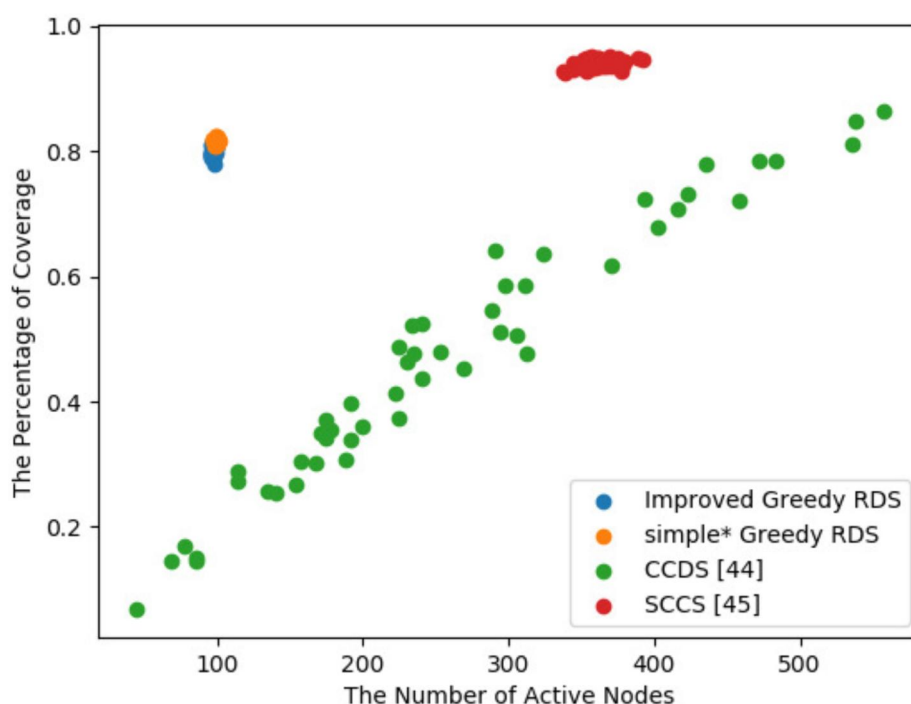


FIGURE 7 The number of active nodes and the percentage of coverage for every random run of different algorithms on a network of size 1200

compared to our proposed method. It also marks more nodes as active; therefore, it leads to more energy consumption (i.e. reducing network lifetime), as the amount of energy consumed by the network is mostly determined the number of its active nodes. However, the CCDS method may have a slightly better performance compared to our method, but it also uses a considerably larger number of active nodes and hence is less energy efficient compared to the proposed method.

6 | CONCLUSIONS

In this study, we have considered maximising the lifetime of a wireless sensor network using a sleep/wake-up schedule while maintaining partial coverage. We model the notion of partial coverage of an area with sensor nodes using Roman domination. We discuss the definition of Roman domination and prove three theorems regarding the minimum Roman dominating number. Afterwards, we present the simple greedy algorithm and an improved greedy algorithm for calculating the minimum Roman dominating number. We then compare the two algorithms in several experiments, where we can notice a considerable improvement in the results of the second algorithm, which means that the second algorithm can considerably improve the lifetime of a wireless sensor network.

Future studies can pursue this research to find a connected Roman dominating set and use it as a backbone for the wireless sensor network. Another line of work would be implementing the proposed model on a WSN with a probabilistic sensing model.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable.

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